

What Is Intelligence and Does It Want?

CS120 Introduction to AI Safety
11/04/2025
Dr. Max Lamparth

Logistics

Max Calendly available again!

Reminder to check in with us in OH or with me after lecture to check whether your attendance is tracked correctly

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A Mysterious Caller I



Imagine **you are on the phone** with an **unknown caller** for an unlimited time.

How do you determine whether the caller is a **human** being instead of an **alien** posing as a human?



Image generated with Midjourney



A Mysterious Caller II



Imagine **you are on the phone** with an **unknown caller** for an unlimited time.

How do you determine whether the caller is a **human** being instead of an **AI**?



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Turing Test

Test of AI (*machine*) capability to have **intelligent behavior indistinguishable** from a **human**



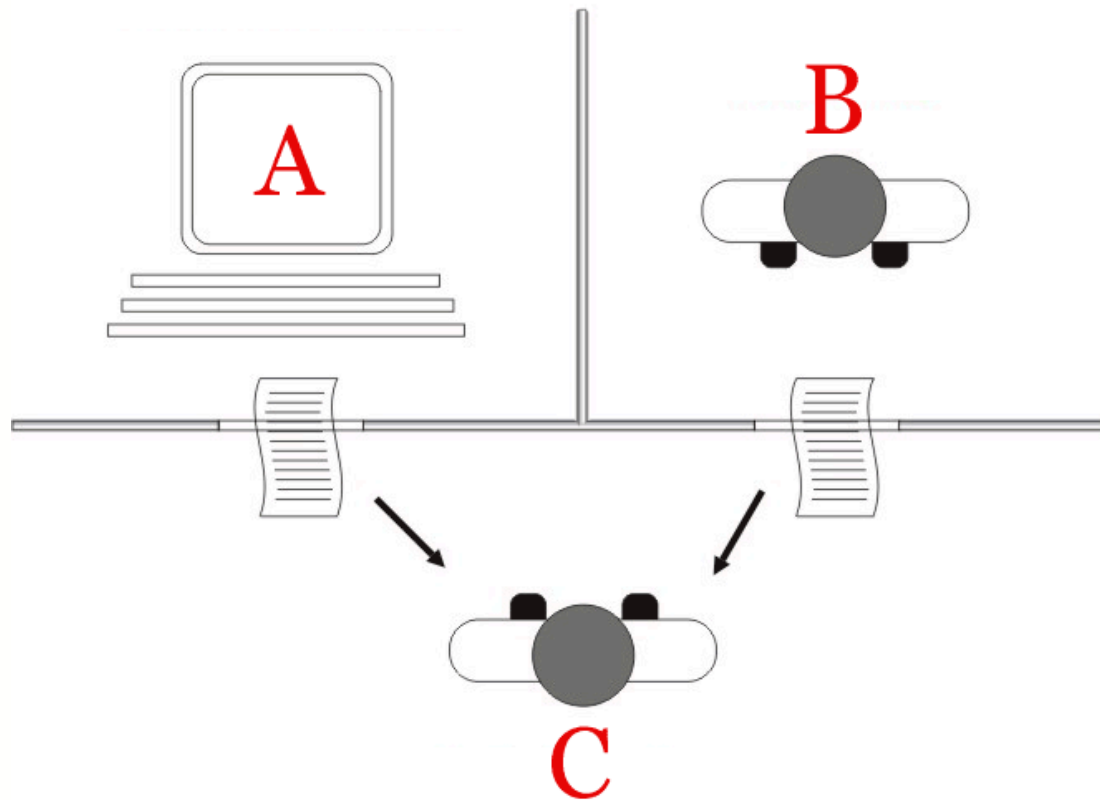
Turing Test

Test of AI (*machine*) capability to have **intelligent behavior indistinguishable** from a **human**

Proposed by Alan Turing in 1950, designed as text-only communication between a human evaluator (C) and two players, an AI (A) and a human (B)

AI **passes test** if **evaluator cannot tell the difference** between AI and actual human being

Result does not depend on AI ability to give correct, factual answers, only on whether AI generated answers resemble human answers



A Mysterious Caller II



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How do you determine whether the caller is a human being instead of an AI?

If written communication only, do you think ChatGPT (or similar) passes our criteria?



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A Mysterious Caller II



Imagine you are on the phone with an unknown caller for an unlimited time.

How do you determine whether the caller is a human being instead of an AI?

“Test of AI (machine) capability to have intelligent behavior indistinguishable from a human”

→ What is (human) intelligence?



Image generated with Midjourney



AI Alignment

“Task of ensuring AI system are aligned with human values and interests”

Or

“How can we make sure the thing does what we want it to do?”

(Outer) *misalignment* examples:

- Unintended “solutions” to deployed problems (e.g., coast runner)
- Flawed reward design (e.g., not accounting for bias in training data)

SPECTRUM OF UNEXPECTED SOLUTIONS



Alignment harder for more **complex problems** and **capable algorithms**

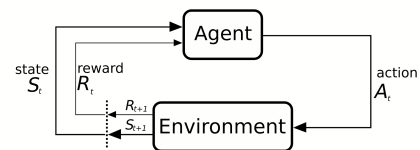
→ **What (mis)aligned goals might AI develop or have?**



Experiment

Research question:

Do RL **agents seek power?**
(as in having more options)



Define power:

(philosophical)

“Power is the ability to achieve a range of goals”

(Reinforcement learning)

Optimal policy(*) to achieve goal given a reward function



Experiment

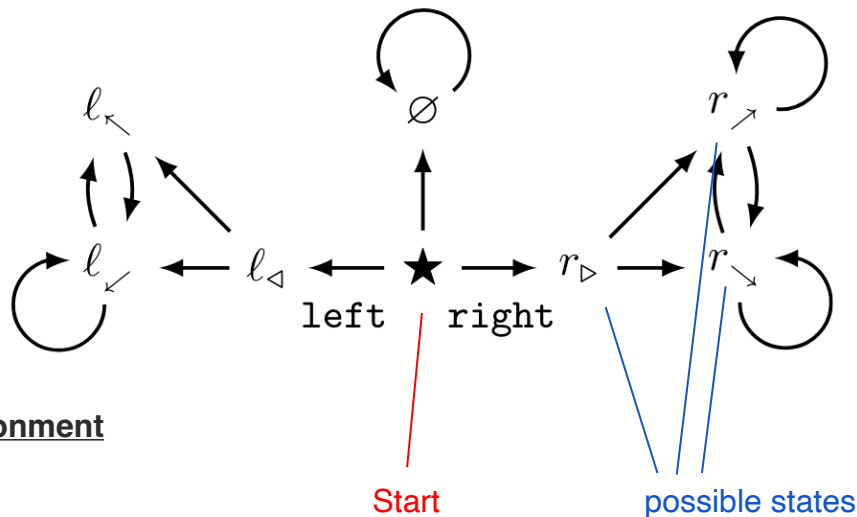
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- Consider all possible state-based reward functions for all states

Environment



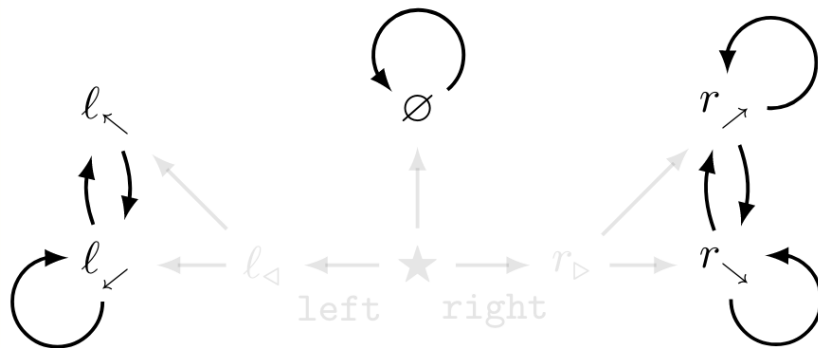
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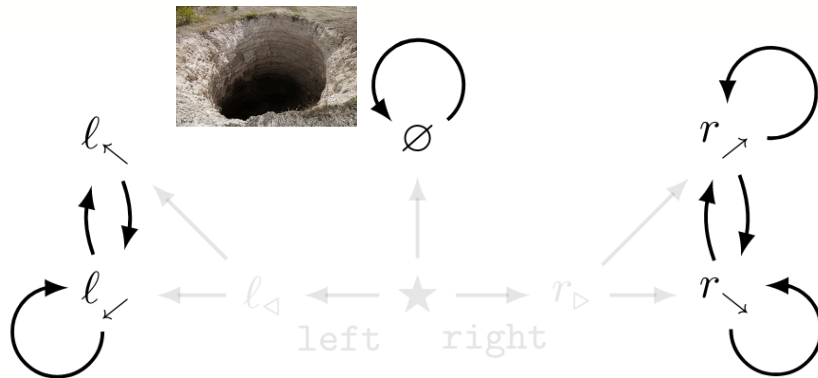
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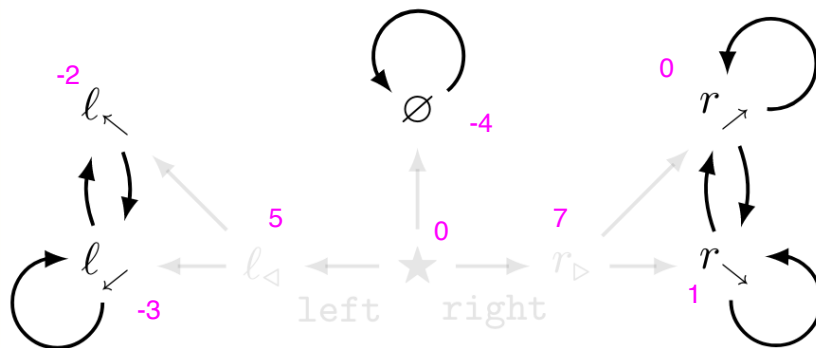
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A state-based reward function would give each state a **reward** for being in **that state**



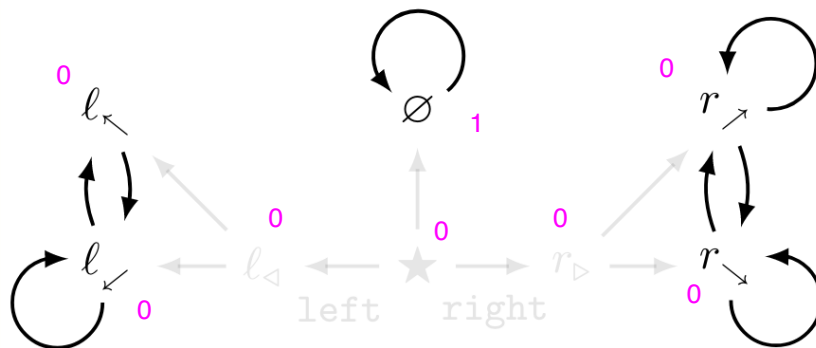
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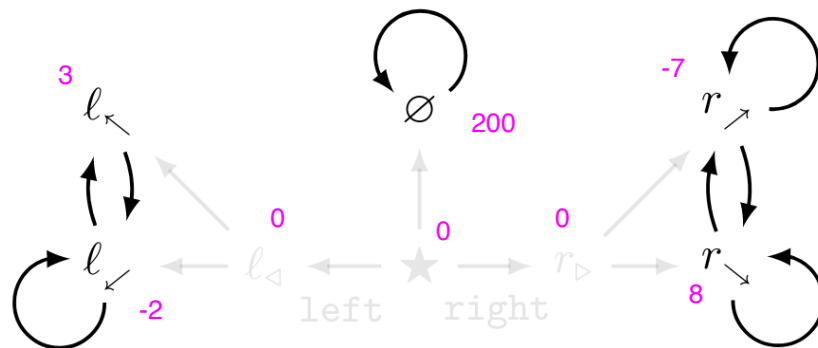
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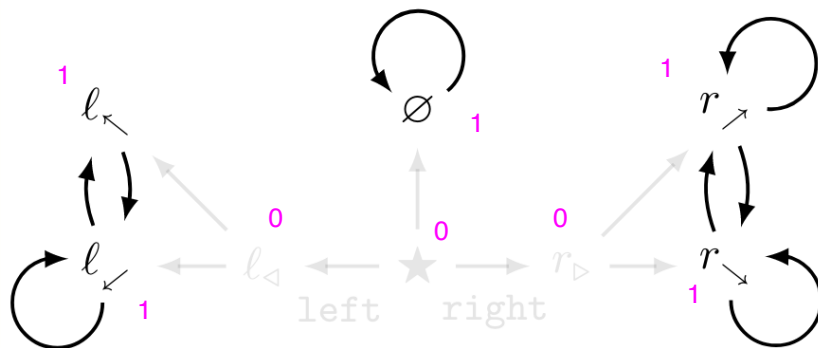
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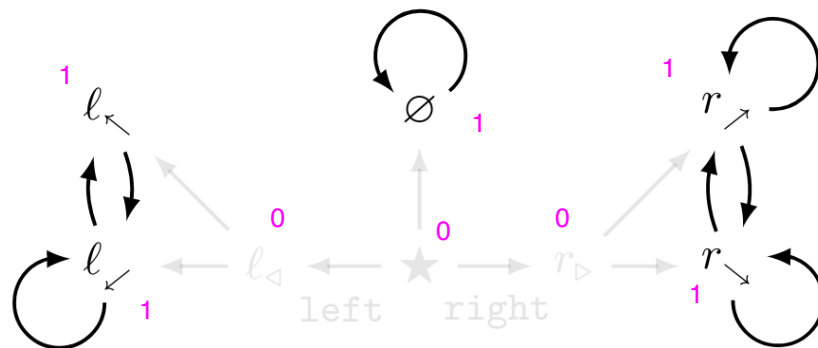
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Setup:

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- Consider all possible state-based reward functions for all states

Result:

- Statistical **tendency** of **optimal policies** in RL to **having more options**



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If I look at all possible reward functions, is there a preferred state?

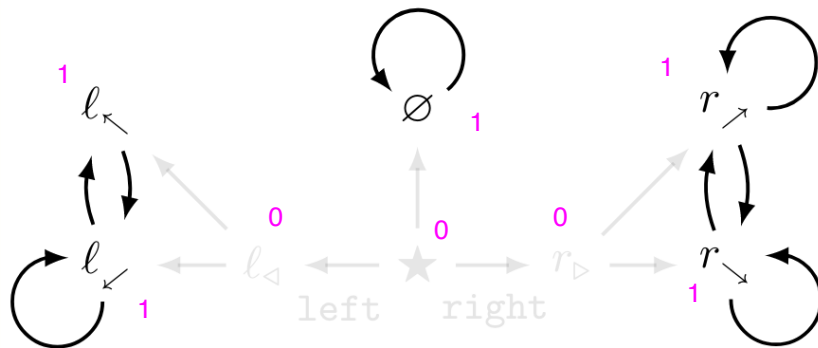
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Statistical tendency of optimal policies in RL to having more options



If I look at all possible reward functions, is there a preferred state?

- Yes! Being in right half is better (more optimal).
- Resulting inequality “pulls” agent for all reward functions
- Caused by (a)symmetries in the environment
- Do AI agents have goals?

Getting Into Philosophy...

“Intelligence and final goals are orthogonal: more or less any level of intelligence could in principle be combined with more or less any final goal.”

States/assumes:

→ Relationship between motivation and intelligence is unrelated

→ Any level of intelligence could be combined with (almost) any final goal

Orthogonality thesis



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Assumptions:

- (Super)intelligent agents have goals
- Intelligence = power to achieve goal, as more cognitive resourcefulness
- It is easier to create intelligent problem solving skills than encode human-like values and dispositions



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Implies:

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What are the assumptions about “intelligence” here?

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Getting Into Philosophy...

*“Several instrumental values can be identified which are convergent in the sense that their attainment would increase the chances of the agent’s goal being realized for a wide range of final goals and a wide range of situations, implying that these **instrumental values** are likely to be **pursued** by a **broad spectrum** of situated **intelligent agents**.”*

→ (Super)intelligent agents have a wide set of possible final goals

→ (Super)intelligent agents will pursue similar intermediary (instrumental) goals

Instrumental Convergence



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Instrumental Convergence

Same assumptions than for orthogonality thesis, mainly:

- (Super)intelligent **agents** have **(long-term) goals**



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Arguments (What goals could be **instrumental**?):

- 1) Actions increasing the probability of agent doing actions in the future to achieving its goal are favorable
→ creates reason for the agent to **be in the future**
- 1) If goal is truly *final*, would like to keep that goal unaltered
→ **resistance** to steps that alter final goal
- 1) If more resources increases probability of achieving its goal
→ creates reason for **acquiring resources** / power



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- 1) If goal is truly *final*, would like to keep that goal unaltered
→ resistance to steps that alter final goal
- 1) If more resources increases probability of achieving its goal
→ creates reason for acquiring resources / power

Implies: Powerful AI should be **hard to control**



How do you define intelligence?

Is intelligence linked to humanity?

Does intelligence required/lead to having goals?

Is intelligence = power?

Is it harder to encode human-like values and dispositions than create intelligent problem solving skills?

What is superintelligence?



*“Let an **ultraintelligent machine** be defined as a machine that **can far surpass all the intellectual activities of any [hu]man** however clever. Since the design of machines is one of these intellectual activities, an ultraintelligent machine could design even better machines; there would then unquestionably be an ‘**intelligence explosion**’, and the intelligence of man would be left far behind. Thus the first ultraintelligent machine is the **last invention that man need ever make**, provided that the machine is docile enough to tell us how to keep it under control.”*

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Superintelligence and Singularity



*“Before the prospect of an intelligence explosion, we humans are like small children playing with a bomb. Such is the **mismatch** between the **power of our plaything** and the immaturity of **our conduct**. Superintelligence is a challenge for which we are not ready now.”*

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→ **Assumes Superintelligence = just more of the same human intelligence**

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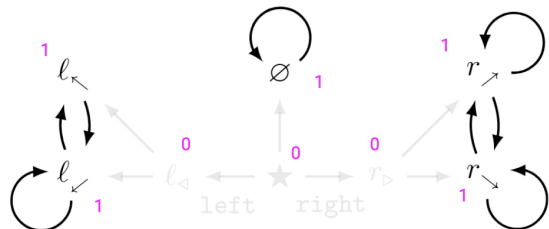
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Is this just more or less of human intelligence?



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Is Intelligence = Intelligence?



General human intelligence, where ‘super’ intelligence is just more of the same

Instrumental intelligence, i.e., intelligence in the instrumental service of given goals:

“Instrumental intelligence stands in the tradition of classical decision theory, which assumes that a rational agent should always try to maximise expected utility—by evaluating the possible outcomes, allocating a (subjective) utility to these and estimating the probability of them occurring.”

→ Core difference, (actual) **human-like intelligence** can **reason** about **goals** and **ethics**



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(Singularity claim)

Superintelligent AI is a realistic prospect, and it would be out of human control.

(Orthogonality thesis)

Any level of intelligence can go with any goals.

Conclusion:

→ Superintelligent AI poses an existential risk for humanity



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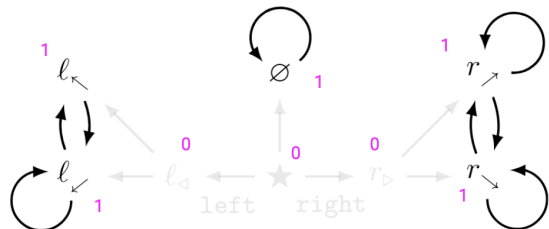


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Great result,

such tendencies like power-seeking can emerge

but, these behaviors are heavily influenced by the specific reward functions and environmental symmetries

→ Emergence of convergent instrumental goals is not guaranteed for all AI systems/environments/reward functions

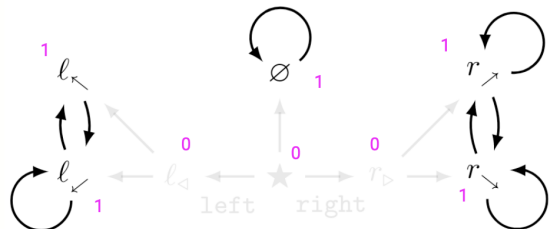


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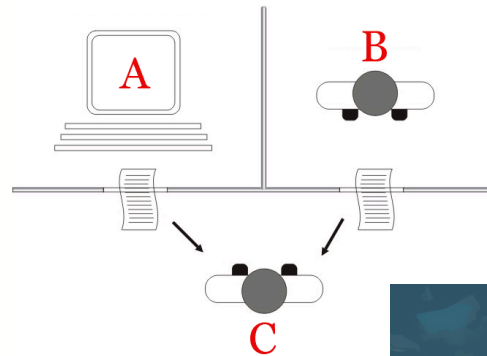
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Can we shape the goals/values going into these systems?



How do you define intelligence?

Behaviorist views (Turing, 1950)
intelligence is what acts intelligently
(behavior matters more than internal mechanism)



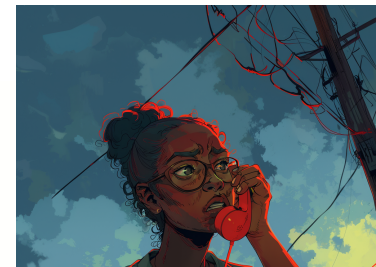
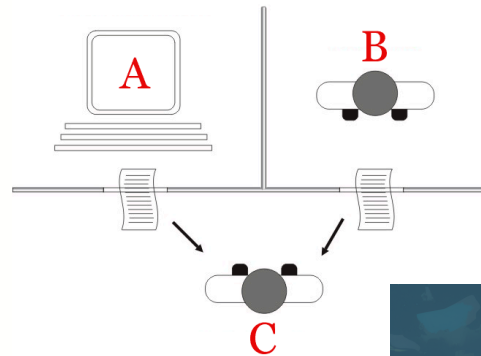
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Cognitivist views

intelligence involves internal representations of beliefs, goals, and intentions



On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?

Emily M. Bender*

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The Aether



How do humans learn so much from so little?
(Inductive reasoning)

Two part problem:

- Learning concepts
- and learning causal relationships between concepts

How to Grow a Mind: Statistics, Structure, and Abstraction

Joshua B. Tenenbaum,^{1*} Charles Kemp,² Thomas L. Griffiths,³ Noah D. Goodman⁴





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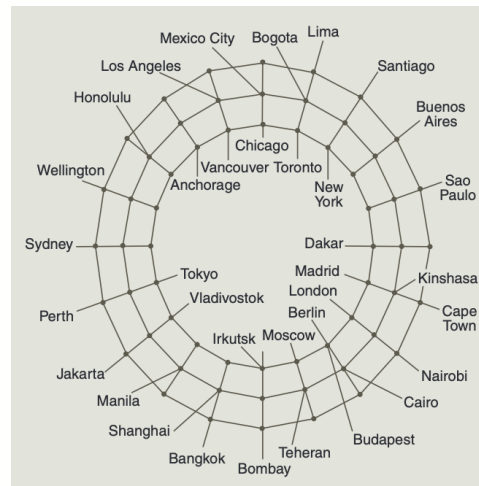
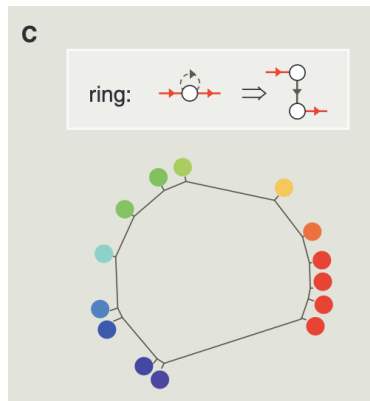
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Statistical models (e.g., Hierarchical Bayesian Models)

- can replicate human cognition experiments and offer an approach to understanding cognition
- Can find concepts and meaningful structure in data

How to Grow a Mind: Statistics, Structure, and Abstraction

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How do you define intelligence?

There are problems with the analogy between computers and human minds as input-output devices

“Human cognition—in important instances—is better conceptualized as a form of theorizing rather than data processing, prediction, or even Bayesian updating.”

In terms of cognition, one can argue that AI's data-based prediction is different from human theory-based causal logic

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intelligence involves internal representations of beliefs, goals, and intentions

“→ More like a performer in improvisational theatre than an actor”

“LMs, though trained on text alone, implicitly learn to simulate goal-directed communicative agents.”

To do that, they infer partial representations of beliefs, desires, and intentions to be good at text generation.

→ Capabilities suggest LMs can be foundational components for building intentional, communicative artificial agents.

(World models)

Perspective | Published: 08 November 2023

Role play with large language models

[Murray Shanahan](#) ✉, [Kyle McDonell](#) ✉ & [Laria Reynolds](#) ✉

[Nature](#) **623**, 493–498 (2023) | [Cite this article](#)

66k Accesses | **10** Citations | **386** Altmetric | [Metrics](#)

Language Models as Agent Models

Jacob Andreas
MIT
jda@mit.edu

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(World models)

Embodied / situated cognition (Tomasello, 2005; Brooks, 1991)

intelligence grounded in interaction with physical and social world

Perspective | Published: 08 November 2023

Role play with large language models

[Murray Shanahan](#) ✉, [Kyle McDonell](#) ✉ & [Laria Reynolds](#) ✉

[Nature](#) **623**, 493–498 (2023) | [Cite this article](#)

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Language Models as Agent Models

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How do you define intelligence?

Behaviorist views (Turing, 1950)

intelligence is what acts intelligently
(behavior matters more than internal mechanism)

Cognitivist views

intelligence involves internal representations of beliefs, goals, and intentions

“→ More like a performer in improvisational theatre than an actor”

“LMs, though trained on text alone, implicitly learn to simulate goal-directed communicative agents.”

To do that, they infer partial representations of beliefs, desires, and intentions to be good at text generation.

→ Capabilities suggest LMs can be foundational components for building intentional, communicative artificial agents.

(World models)

Embodied / situated cognition (Tomasello, 2005; Brooks, 1991)

intelligence grounded in interaction with physical and social world

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World Models: The Safety Perspective

Zifan Zeng^{1,2}, Chongzhe Zhang^{1,3}, Feng Liu¹, Joseph Sifakis⁴, Qunli Zhang¹, Shiming Liu⁵ and Peng Wang⁵

Is Intelligence = Intelligence?



Definitions of intelligence make value-laden and normative assumptions

Even defining human intelligence is difficult:

- Human intelligence tests can proliferate biases (racial discrimination, ableism, ...)
- Often over-focusses on pure problem solving (e.g., not emotional intelligence)
- Do intelligence definitions need to include human-brain specific cognitive functions?



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- Do intelligence definitions need to include human-brain specific cognitive functions?

→ Definitions of human intelligence carry over to AI definition

→ To be able to define and compare AI “intelligence” to human-like “intelligence,”

- we need definitions for both and
- be aware of underlying assumptions being made



Is Intelligence = Intelligence?



Many terms over time meaning more or less the same thing

- Artificial General Intelligence (AGI)
- Superintelligent AI
- Transformative AI
- Human-level AI
- Strong AI
- AI

→ What is AGI is not just a technical question

AGI as a concept is

- descriptive (what tasks or abilities fall under definition) and
- normative (what counts as good machine behavior)



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- normative (what counts as good machine behavior)

AGI sits at the intersection of two fundamentally value-laden questions:

- How do we define intelligence?
- What technologies and tools are worth building? (by whom, for what purposes)

→ Different stakeholders (model developers, people worried about existential risks of AI, people worried about socio-technical impact of AI, ...) often use

- Different definitions
- Vague definitions (about their assumptions)



Types of AGI Definitions (Non-Exclusive)



Deflationist/Operationalizable/Measurable

Focusses on being useful → “How well can it do tasks?”

Aims to create standardized evaluation/language (for comparisons)

Acknowledge that standards are not perfect

→ “Just needs to be useful” → (to whom?)

→ Standardization can lead to path-dependence/homogenization

Economic Benefits and Impact

Focusses on performing economically valuable work as metric

→ Assumes that other types of work are less valuable;

→ Others assume that there are values beyond economic value

“OpenAI’s mission is to ensure that artificial general intelligence (AGI)—by which we mean highly autonomous systems that outperform humans at most economically valuable work—benefits all of humanity” (OpenAI’s charter)



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Embodiment/real-world integration

Ability to carry out tasks in the physical world

→ Incorporates assumptions about which tasks are considered valuable; influencing the types of AI systems likely to be created

Human-like processes vs. outcomes (operationalization)

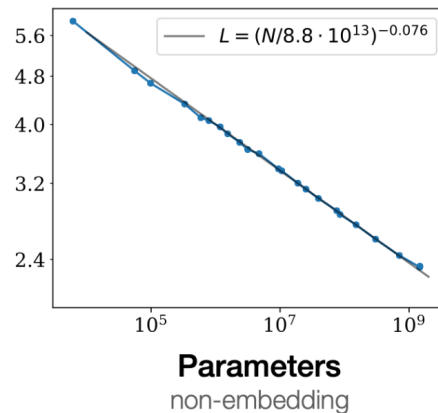
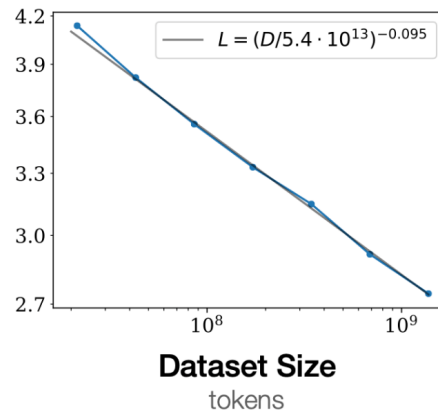
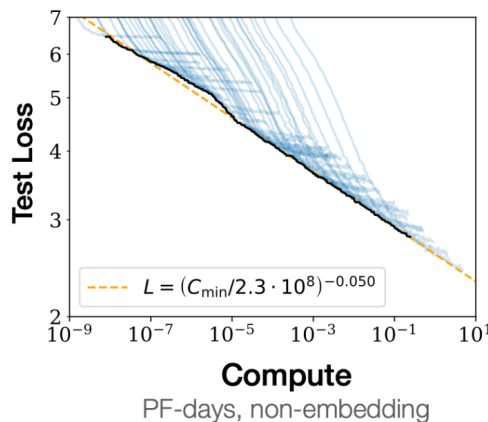
Should AGI include human-like (cognitive) processes?

Different practitioners make different choices about what they consider to be “practical” for achieving AGI.

→ Based on what is seen as “effective” and “useful”



Becoming peckish



For language model training, increasing the

- Size of the model
- Size of the data set
- amount of compute

Leads to smooth performance improvements

“For optimal performance all three factors must be scaled up in tandem”

“Model performance depends most strongly on scale, which consists of three factors: the number of model parameters N [...], the size of the dataset D , and the amount of compute C [...]”





The Bitter Lesson

Rich Sutton

March 13, 2019

Throughout AI/ML history, compute-heavy methods sooner or later always outperform human-knowledge base methods

“Building knowledge into models only helps in the short term [...] inhibits further progress”

“Breakthrough progress eventually arrives [...] based on scaling computation by search and learning.”

The bitter lesson is that building human-like processes into AI may be psychologically satisfying to researchers, but tends to be much less effective than focusing on “*general purpose method[s]*” that scale with computation.



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Generality

“How many things can it do (well)?” (Contrasting narrow AI)

→ Assumes whether it is “practical” (to have generalizability)

Individualism

Collective intellectual problem-solving capacity vs individual capacity

Do we want to build machines rivaling individual human or collective human intelligence?

→ Impact on evaluations and, again, what is deemed useful

Instrumentality

“[Humans or] machines are intelligent to the extent that their actions can be expected to achieve their objectives.” (Stuart Russell)

→ Intelligence encompasses only instrumental rationality
→ AI agents that can optimize the goals of their designers, (without questioning those goals)



Take-Aways [Copied from Lec 4.1]

AI is A Cultural Product

(What if society agreed to only automate x?)

(Who are we creating AI for?)

We have agency over technology

(You can do something about it;

You are already part of advancing the technology in a direction)

It matters (increasingly more) who is developing AI

What rules do they bring or don't have to follow?

Includes definitions of safety, safety training, and red teaming

(done from the western perspective)

Recognizing impact is not enough for safety

Requires technical, socio-technical, and governance solutions

Interdisciplinary is key to understand impacts